EE610 – Assignment 2

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22B1216

Question 1 – Noise modelling and histogram matching

(a)(i) Discretising a Gaussian noise model

A continuous Gaussian density with mean μ_n and variance σ_n^2 is sampled at the integer offsets $k \in \{0, 1, \ldots, 255\}$ around a shift $s = 127$ (so $k = 127$ corresponds to zero noise) and renormalised to a PMF:

$$p_n(k) \;=\; \frac{1}{Z}\,\exp\!\left(-\frac{(k-s-\mu_n)^2}{2\sigma_n^2}\right), \qquad Z = \sum_{k=0}^{255} \exp\!\left(-\frac{(k-s-\mu_n)^2}{2\sigma_n^2}\right).$$

Samples are drawn by inverse-transform sampling: compute the CDF $F(k) = \sum_{j \le k} p_n(j)$, draw $u \sim \mathrm{Uniform}(0,1)$, and pick the smallest k for which $F(k) \ge u$. The noise pixel value is then $k - s$.

(a)(ii) Sample mean and variance of $g = f + n$

With n independent of f and $\mathbb{E}[n] = 0$,

$$\mathbb{E}[g] \;=\; \mathbb{E}[f] + \mathbb{E}[n] \;=\; \mathbb{E}[f], \qquad \mathrm{Var}(g) \;=\; \mathrm{Var}(f) + \mathrm{Var}(n) + 2\,\mathrm{Cov}(f,n) \;=\; \mathrm{Var}(f) + \sigma_n^2.$$

The covariance term vanishes by independence. (Empirically the equality is approximate because clipping to $[0, 255]$ introduces a small bias.)

(b) Implementation and (c) Verification

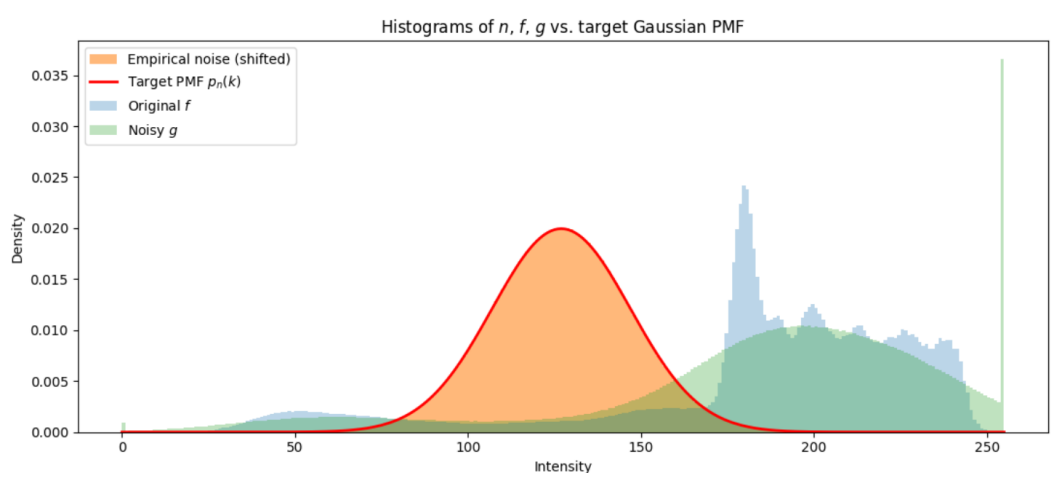

Figure 1: Empirical histograms of n (shifted by $+127$), f, and g overlaid on the target PMF $p_n(k)$.

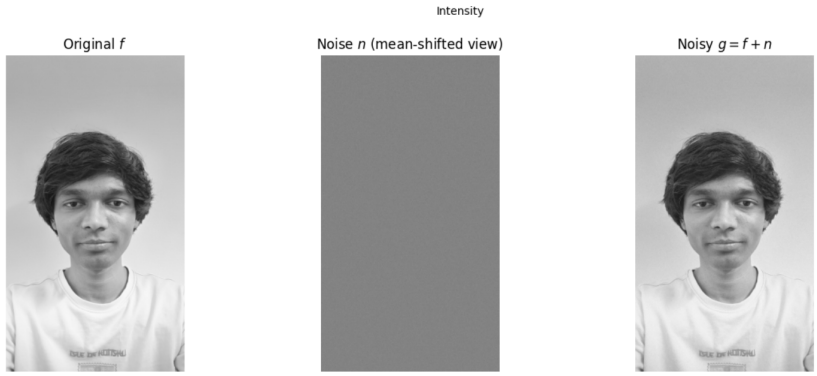

Figure 2: Original image f and degraded image $g = f + n$.

Question 2 – Importance of phase

(a) Conceptual justification

From the inverse 2-D DFT,

$$f(x,y) \;=\; \frac{1}{MN}\sum_{u,v} |F(u,v)|\,\exp\Big(j\big[2\pi(ux/M + vy/N) + \phi(u,v)\big]\Big).$$

$|F(u,v)|$ controls how much of each frequency is present, while $\phi(u,v)$ controls the spatial alignment of those frequencies. Sharp edges and object boundaries form where many spectral components add constructively at the same location – that alignment is encoded entirely in the phase. Hence $g_A = \mathcal{F}^{-1}\{|F_1|\,e^{j\phi_2}\}$ inherits the geometry of image 2 and the "colour palette" of image 1.

(b) Spectra and reconstructions

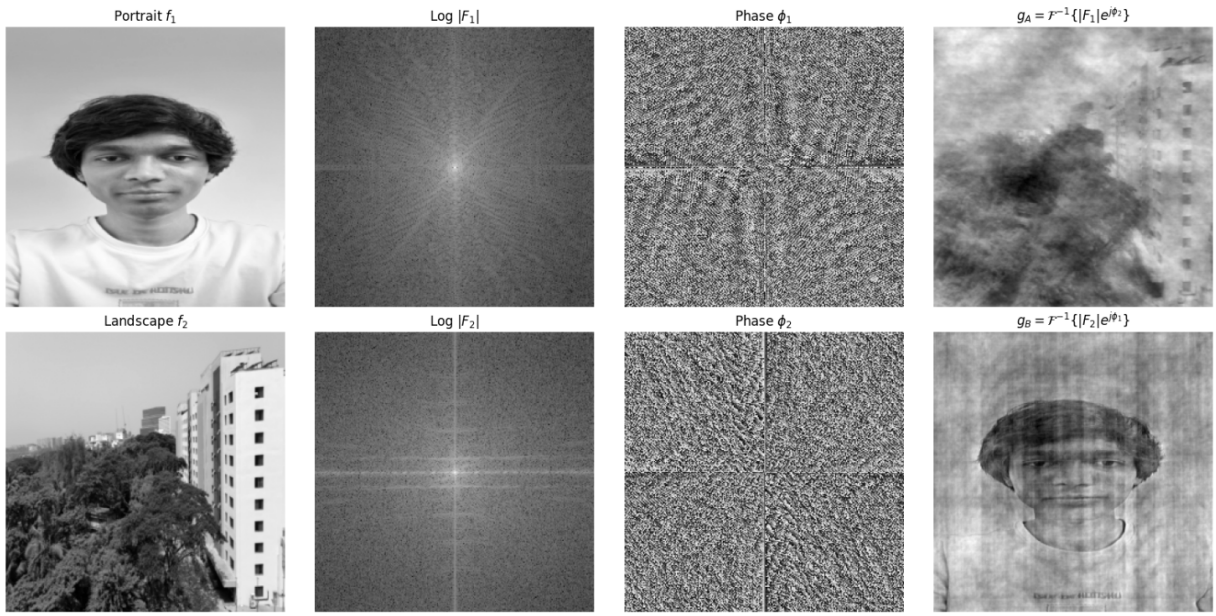

Figure 3: Top: portrait, $\log|F_1|$, phase of F_1, and g_A. Bottom: landscape, $\log|F_2|$, phase of F_2, and g_B.

Question 3 – Radon transform and filtered backprojection

(a)(i) Radon transform

For an angle θ and offset ρ, lines are parametrised by $\rho = x\cos\theta + y\sin\theta$, and

$$R(\rho,\theta) \;=\; \iint f(x,y)\,\delta(x\cos\theta + y\sin\theta - \rho)\,dx\,dy.$$

Stacking $R(\rho,\theta)$ over $\theta \in [0°, 180°)$ produces the sinogram.

(a)(ii) Fourier-slice theorem

The 1-D Fourier transform of a projection $R(\cdot,\theta)$ equals a radial slice of the 2-D Fourier transform of f at angle θ: $\mathcal{F}_\rho\{R(\rho,\theta)\}(\omega) = F(\omega\cos\theta,\, \omega\sin\theta)$. Plain backprojection accumulates these slices in the spatial domain. Because more slices pass near the origin than near the periphery, the low frequencies are oversampled by a factor $1/|\omega|$, producing the characteristic blur ("halo"). Multiplying each projection by a ramp $|\omega|$ in the frequency domain compensates for this bias – this is filtered backprojection.

(b)(c)(d) Implementation

Bilinear interpolation is used inside the rotation routine. The 1-D FFT/IFFT are implemented from scratch (iterative Cooley-Tukey). Each projection is zero-padded to twice the next power of two, multiplied by a ramp×Hamming filter, inverse-FFT'd, and smeared back across the image after rotation by $-\theta$.

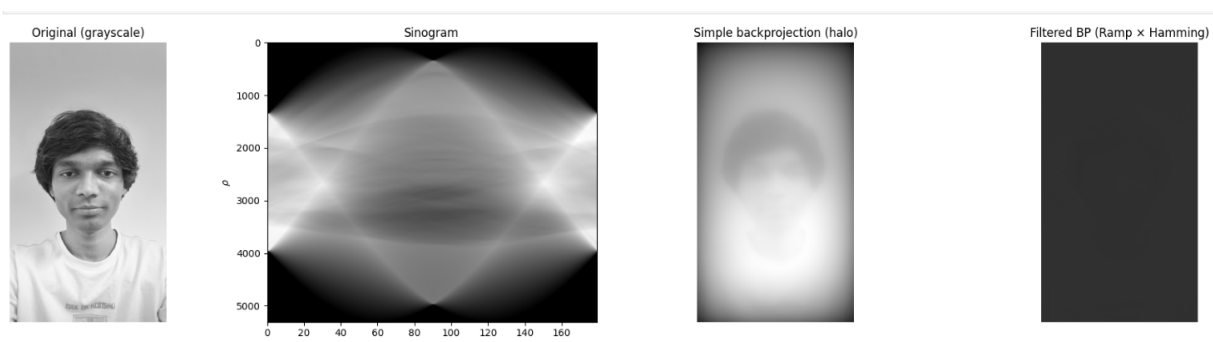

Figure 4: Original image, sinogram (1° angular step), simple backprojection, and filtered backprojection (ramp × Hamming).

Question 4 – 2-D sinusoidal gratings

(a) Roles of the parameters

- A: amplitude / contrast of the stripes (peak-to-peak intensity).
- u_0 alone (with $v_0 = 0$): vertical stripes; u_0 stripes fit across the image width, so the period is M/u_0 pixels.
- u_0 and v_0 together: tilted stripes. Orientation is $\arctan\left(\frac{v_0 M}{u_0 N}\right)$ and the spatial frequency magnitude is $\sqrt{(u_0/M)^2 + (v_0/N)^2}$.

(b) Why two impulses

A real cosine is the sum of two complex exponentials of equal amplitude at frequencies $\pm(u_0, v_0)$: $\cos\theta = \frac{1}{2}e^{j\theta} + \frac{1}{2}e^{-j\theta}$. Hermitian symmetry of any real-valued image forces $F(-u, -v) = F^*(u, v)$, so a single cosine produces exactly two symmetric impulses at (u_0, v_0) and $(-u_0, -v_0)$.

(c) Varying u_0 with $v_0 = 0$

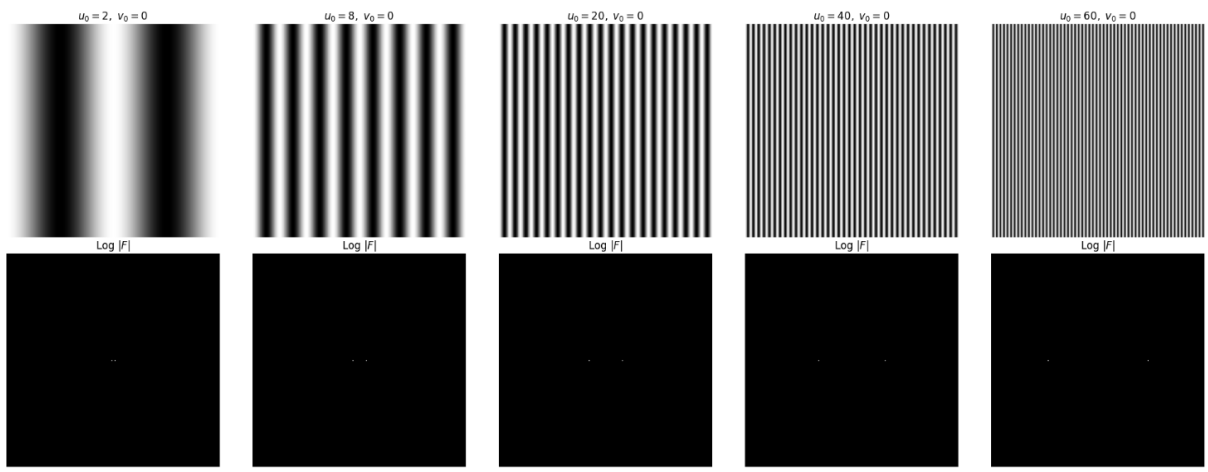

Figure 5: Five gratings with $u_0 \in \{2, 8, 20, 40, 60\}$, $v_0 = 0$, and their log-magnitude spectra.

(d) Constant total frequency, varying orientation

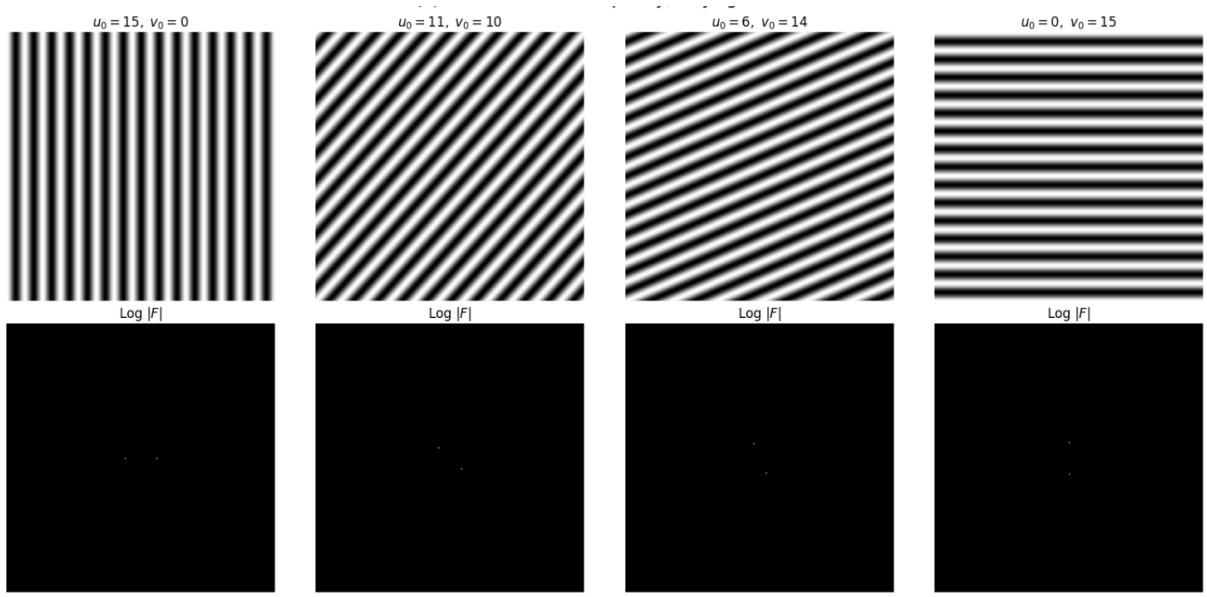

Figure 6: Four gratings with $\sqrt{u_0^2 + v_0^2} \approx 15$ and their spectra.

Question 5 – Adaptive and bilateral filtering

(a) Noise generation

Additive Gaussian noise with $\sigma = 20$ is added to the image and clipped to $[0, 255]$.

Figure 7: Original, Gaussian noise ($\sigma = 20$), and degraded image.

(b) Adaptive local noise-reduction filter

On a 5×5 window centred at (x, y),

$$\hat{f}(x, y) \;=\; g(x, y) - \frac{\sigma_n^2}{\sigma_L^2}\left[g(x, y) - m_L\right], \qquad \sigma_L^2 \leftarrow \max(\sigma_L^2, \sigma_n^2),$$

so that the variance ratio is capped at 1 in noise-dominated regions and approaches 0 at sharp edges (preserving them).

(c) Bilateral filter

$$I'(x) \;=\; \frac{1}{W(x)} \sum_{y \in S} I(y)\, \exp\!\left(-\frac{\|x - y\|^2}{2\sigma_s^2}\right) \exp\!\left(-\frac{|I(x) - I(y)|^2}{2\sigma_r^2}\right),$$

with $\sigma_s = 3,\ \sigma_r = 30$ and a 13×13 window.

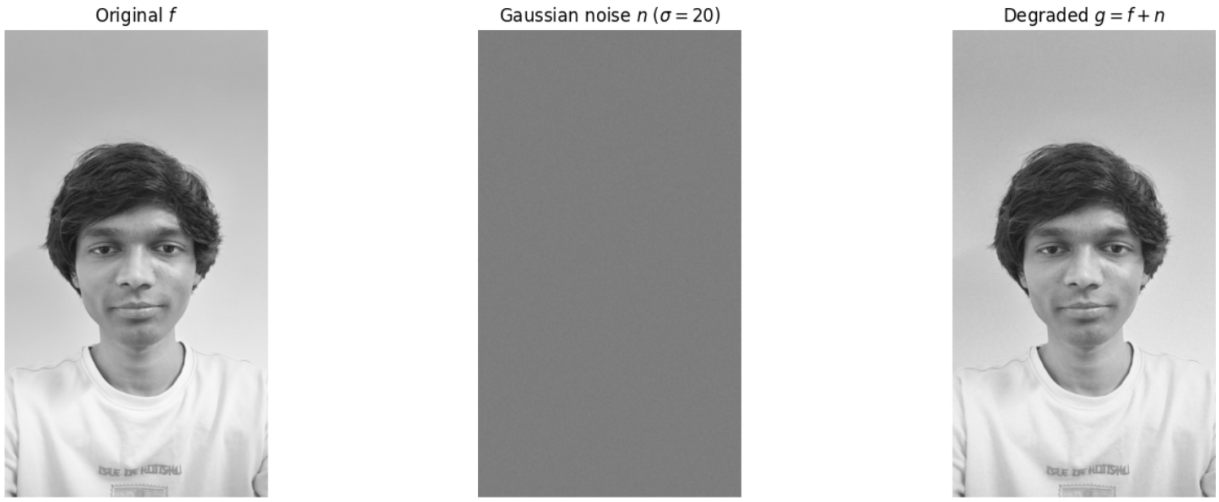

Figure 8: Degraded image, adaptive local noise reduction, and bilateral filter outputs.

Question 6 – Wiener restoration of motion blur

(a)(b) Degradation model

A motion-blur kernel

$$H(u,v) \;=\; \frac{T}{\pi(ua+vb)}\sin\big(\pi(ua+vb)\big)\,e^{-j\pi(ua+vb)}, \qquad H(u,v)=T \text{ when } ua+vb=0,$$

with $a = b = 0.05$, $T = 1$, is applied in the frequency domain. Zero-mean Gaussian noise of variance $\sigma_n^2 = 100$ is added and the result is clipped to $[0, 255]$.

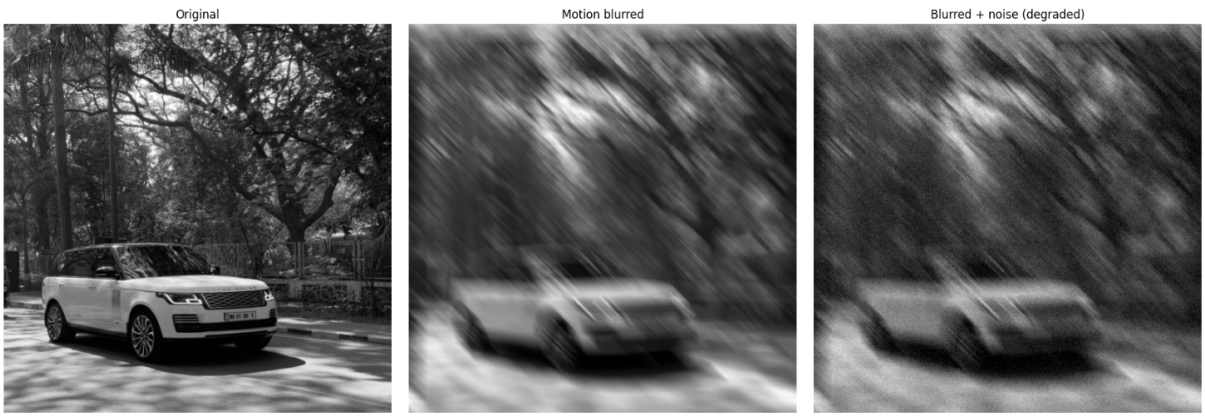

Figure 9: Original, motion-blurred, and degraded (blur+noise) images.

(c) Direct inverse filtering

$\hat{F}(u,v) = G(u,v)/H(u,v)$ amplifies noise wherever $|H|$ is small.

(d) Wiener filter

$$\hat{F}(u,v) \;=\; \left[\frac{1}{H(u,v)}\cdot\frac{|H(u,v)|^2}{|H(u,v)|^2+K}\right]G(u,v).$$

Figure 10: Direct inverse filter and Wiener filter outputs for $K \in \{0.001, 0.01, 0.05, 0.1\}$.

Method	K	MSE
Direct inverse filter	—	`21700.14`
Wiener filter	0.001	`7243.02`
	0.01	`1130.30`
	0.05	`732.32`
	0.1	`837.38`

Table 1: MSE values for inverse filtering and Wiener filtering.